# 7B+5C
# STOICHIOMETRY

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

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# LEARNING OBJECTIVES

## CHAPTER 7

1. Define gas pressure and its units.
2. Calculate the number of moles, pressure, volume, or temperature in a sample of gas using the quantities of the other three.
3. Calculate the properties of each gas in a mixture of gases.
4. Predict how the pressure in a fixed-volume container will change after a complete reaction.
5. Use the kinetic molecular theory to explain the behavior of gases.
6. Describe the relationship between molecular mass and speed.
7. Calculate the relative speeds and molecular masses of gases using root mean square speed and Graham's law of effusion.

# Gases in Chemical Reactions

7.9

## Gaseous stoichiometry

- Recall, stoichiometry is *always* about moles.
- For gases, use the ideal gas law to determine the moles of gases present.

# IN-CLASS QUESTION 1

Learning Objective(s): 7.4

A 2.00 g sample of aluminum metal is combined with 1.50 L of oxygen gas at 20.0 °C and 2.25 atm. **What mass (in grams) of solid aluminum oxide can be formed?**

# IN-CLASS QUESTION 2

Learning Objective(s): 7.4

$N_2O_5$ is an unstable gas that decomposes: $2\ N_2O_5(g) \rightarrow 4\ NO_2(g) + O_2(g)$

A sample of $N_2O_5$ is placed into a 5.00 L container at 25.0 °C and an initial pressure of 0.400 atm. After the entire $N_2O_5$ sample decompose, the temperature is 37.2 °C. **What is the partial pressure of $NO_2$ ($P_{NO_2}$ in atm) in the container after decomposition?**

# EXAMPLE PROBLEM

Learning Objective(s): 5.3

Each question statement below is a stoichiometry problem. For each, determine if you would first need to identify a limiting reactant (LR) or not.

$$P_4 + 5\ O_2 \rightarrow P_4O_{10}$$

Burning phosphorus in oxygen produces tetraphosphorus decoxide. What is the percent yield of the reaction if 10.0 g of $P_4O_{10}$ are obtained at the end of the reaction between 0.200 mol $P_4$ and 0.200 mol $O_2$?

- ❑ Yes, I would need to identify a LR.
- ❑ No, I do not need to identify a LR.

What mass (in grams) of $MgCl_2$ is needed to react completely with 10.0 g of $AgNO_3$?

$$MgCl_2(aq) + 2\ AgNO_3(aq) \rightarrow 2\ AgCl(s) + Mg(NO_3)_2(aq)$$

- ❑ Yes, I would need to identify a LR.
- ❑ No, I do not need to identify a LR.

Consider the reaction between 10.0 g Al and 10.0 g $Br_2$ according to the unbalanced equation. If 6.97 g of $AlBr_3$ is obtained in the lab, what is the percent yield?

$$Al + Br_2 \rightarrow AlBr_3$$

- ❑ Yes, I would need to identify a LR.
- ❑ No, I do not need to identify a LR.

# EXAMPLE PROBLEM

Learning Objective(s): 5.2

What mass (in grams) of oxygen is required to produce 27.5 g of $NO_2$ according to the *unbalanced* chemical equation below?

$$\_\_\_ NH_3 + \_\_\_ O_2 \longrightarrow \_\_\_ NO_2 + \_\_\_ H_2O$$

# EXAMPLE PROBLEM

Learning Objective(s): 5.3

If 42.5 g of $O_2$ reacts with 75.1 g of Mg to produce MgO, then what mass (in grams) of the *excess* reagent remains?

# EXAMPLE PROBLEM

Learning Objective(s): 5.5

A 20.00 mL solution of $H_2SO_4$ of unknown concentration reacts with 0.100 M NaOH.

You find that 49.22 mL of 0.110 M NaOH is required for *complete* neutralization (100 % yield). What is the initial concentration (in molarity, M) of the original $H_2SO_4$ solution?